

Biomechanics: Meaning

- Bio: Living organism
- Mechanics: Branch of physical science dealing with forces acting on a body (static or moving).
- Biomechanics: The study of forces and their effects on human beings in moving or static conditions.

Newton's Laws of Motion & Application in Sports

Sir Isaac Newton's three laws describe the motion of massive bodies and their interactions.

1. First Law: Law of Inertia

- Definition: An object remains in its state of rest or uniform motion unless an external force acts upon it.
- Application in Sports:
 - A moving football slows down and stops due to friction.
 - A basketball shot follows a parabolic path due to gravity and slows down due to air resistance.

2. Second Law: Law of Acceleration

- Definition: The rate of change of acceleration is directly proportional to the force applied and inversely proportional to the object's mass ($F = ma$).
- Application in Sports:
 - Hitting a baseball with double the force doubles its acceleration.
 - Applying force to a tennis ball to accelerate it.

3. Third Law: Law of Action and Reaction

- Definition: For every action, there is an equal and opposite reaction.
- Application in Sports:
 - Swimming: Swimmer pushes water backward, water pushes swimmer forward.
 - Running: Runner pushes ground backward, ground pushes runner forward.

Types of Levers & Application in Sports

Levers in the body consist of:

1. Rigid structure: Bone
2. Force: Muscle (produces angular motion)
3. Fulcrum: Fixed point (joint)
4. Load/Resistance: Weight of body part/object being moved

To remember lever types, use FLE (Fulcrum, Load, Effort in the middle).

1. First Class Lever

- Order: Fulcrum in the middle (Effort - Fulcrum - Load).
- Example: Raising the head to head a football. Neck muscles (effort), neck (fulcrum), head (load).

2. Second Class Lever

- Order: Load in the middle (Fulcrum - Load - Effort).
- Example: Standing on tiptoes. Ball of foot (fulcrum), body weight (load), gastrocnemius muscle (effort). Used in jumps or sprint starts.

3. Third Class Lever

- Order: Effort in the middle (Fulcrum - Effort - Load).
- Example: Biceps curl. Elbow joint (fulcrum), biceps (effort), forearm/weight (resistance).

Equilibrium, Center of Gravity & Application in Sports

Equilibrium

- Definition: A state of balance where opposite forces cancel out, and no changes occur.
- Types:
 - Dynamic Equilibrium: Balance of the body during movement (e.g., cartwheel in gymnastics, ball moving at unknown speed).
 - Static Equilibrium: Balance of the body during rest or in a stable position.

Principles to Determine Degree of Stability

1. Broader the base, greater the stability.
2. Lower the center of gravity, higher the stability.
3. Body weight is directly proportional to stability.

Center of Gravity (COG)

- Definition: The point around which an object's mass or weight is evenly distributed or balanced, and through which the force of gravity acts.
- Application:
 - Sprinter in starting blocks: COG aligned along rib cage and forward knee maximizes forward propulsion.
 - See-Saw, Tug of War.

Friction & Sports

- Definition: The force that opposes the motion of a solid object over another.

Types of Friction

1. Static Friction: Acts between surfaces at rest relative to each other (e.g., skiing against snow, lamp on table).
2. Sliding Friction: Resistance between two objects sliding against each other (e.g., block across floor, cards sliding).
3. Rolling Friction: Resists the motion of a ball or wheel; weakest type (e.g., log rolling, vehicle wheels).
4. Fluid Friction: Exists between layers of fluid moving relative to each other (e.g., ink flow in pens, swimming).

Projectile in Sports

- Definition: Any object thrown into space upon which the primary acting force is gravity. Other forces (e.g., air resistance) have minimal effect.
- Trajectory: The path followed by a projectile.
- Example: A batted or thrown baseball.

Factors Affecting Projectile Flight Path

1. Gravity
2. Air Resistance
3. Speed of Release (Initial Velocity)
4. Angle of Release (Maximum range obtained at 45 degrees)
5. Height of Release

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